

Stem Cells and Regenerative Medicine

Context-Dependent Modification of PFKFB3 in Hematopoietic Stem Cells Promotes Anaerobic Glycolysis and Ensures Stress Hematopoiesis

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Abstract

Metabolic pathways are plastic and rapidly change in response to stress or perturbation. Current metabolic profiling techniques require lysis of many cells, complicating the tracking of metabolic changes over time after stress in rare cells such as hematopoietic stem cells (HSCs). Here, we aimed to identify the key metabolic enzymes that define metabolic differences between steady-state and stress conditions in HSCs and elucidate their regulatory mechanisms. Through quantitative ¹³C metabolic flux analysis of glucose metabolism using high-sensitivity glucose tracing and mathematical modeling, we found that HSCs activate the glycolytic rate-limiting enzyme phosphofructokinase (PFK) during proliferation and oxidative phosphorylation (OXPHOS) inhibition. Real-time measurement of adenosine triphosphate (ATP) levels in single HSCs demonstrated that proliferative stress or OXPHOS inhibition led to accelerated glycolysis via increased activity of PFKFB3, the enzyme regulating an allosteric PFK activator, within seconds to meet ATP requirements.

Furthermore, varying stresses differentially activated PFKFB3 via PRMT1-dependent methylation during proliferative stress and via AMPK-dependent phosphorylation during OXPHOS inhibition. Overexpression of *Pfkfb3* induced HSC proliferation and promoted differentiated cell production, whereas inhibition or loss of *Pfkfb3* suppressed them. This study reveals the flexible and multilayered regulation of HSC metabolism to sustain hematopoiesis under stress and provides techniques to better understand the physiological metabolism of rare hematopoietic cells.

Key Points

- Combined isotope tracing, mathematical modeling, and single cell ATP analysis enable high-resolution evaluation of blood cell metabolism.
- Under stress, HSCs quickly accelerate glycolysis to meet ATP demands and maintain hematopoiesis via context-dependent PFKFB3 activation.

eLife assessment

This **important** study provides novel strategies to overcome certain limitations when investigating the metabolism of hematopoietic stem cells, mainly due to their low abundance. The study provides some **solid** evidence suggesting that hematopoietic stem cells mainly use glycolysis (rather than mitochondrial OXPHOS or TCA cycle) as their primary energy source, when proliferating in response to hematopoietic stress, as opposed to homeostasis. With the evidence further strengthened by more direct links between metabolic features and cell proliferation, and after consideration of alternative energy sources, this study would be of great interest to stem cell biologists.

Introduction

Activities governing nutrient requirements and metabolic pathways in individual cells maintain tissue homeostasis and respond to stress through metabolite production. Adenosine triphosphate (ATP), produced via cytosolic glycolysis and mitochondrial oxidative phosphorylation (OXPHOS), is the universal energy currency of all organisms; it regulates all anabolic or catabolic cellular activities^{(1)–(3)}. Precise control of intracellular ATP concentrations is critical, as ATP is the rate determiner of many ATP-dependent biochemical reactions^{(4)–(9)}.

Hematopoietic stem cells (HSCs) are tissue stem cells at the apex of the hematopoietic hierarchy; their function is maintained throughout life by a rigorous metabolic program and a complex interplay of gene expression, epigenetic regulation, intracellular signaling, chromatin remodeling, autophagy, and environmental factors^{(10)–(14)}. Conventional analyses of the metabolic programs of hematopoietic stem and progenitor cells (HSPCs) have revealed diverse differentiation potentials and cell-cycling statuses and coordinated activities that maintain hematopoiesis^{(15)–(26)}. Among the HSPC fractions, HSCs possess unique cell cycle quiescence, high self-renewal and differentiation capacity in response to stimuli, and resistance to cellular stress, including reactive oxygen species and physiological aging^{(10), (27)–(30)}. These properties are regulated by a balance between glycolysis and mitochondrial

OXPHOS, requiring biosynthesis of ATP and various macromolecules that confer resilience to stress⁽³¹⁾. Among the known regulators of ATP-producing pathways, glycolytic enzymes maintain HSCs and hematopoietic progenitor cells (HPCs) by regulating cellular survival and cell cycle quiescence^{(32)–(34)}. Loss of mitochondrial genes in HPSCs also induces HSC differentiation defects^{(35)–(37)}. Moreover, disrupting the mitochondrial complex III subunit depletes both differentiated hematopoietic cells and quiescent HSCs⁽²¹⁾. Although glycolysis and the tricarboxylic acid (TCA) cycle are metabolically linked, pyruvate dehydrogenase kinase activity, which can uncouple these pathways, is required to maintain HSC function^{(32), (38)}.

During HSC division, cell metabolism is reprogrammed to activate fatty acid β -oxidation and purine metabolism^{(39)–(41)}. Furthermore, Liang et al. reported that activated HSCs mainly rely on glycolysis as their energy source⁽⁴²⁾. However, the mechanisms by which each ATP-producing pathway and their connections are differentially regulated between HSCs and differentiated cells at steady state, during cell cycling, or during stress remain unknown. Recently, it has been shown that deeply quiescent HSCs do not activate cell cycle under stress^{(43)–(45)}. Therefore, it remains unclear whether metabolic changes such as the individual ATP-producing pathways and their interconnections occur uniformly in all HSCs, including these deeply quiescent HSCs. Furthermore, the underlying hub metabolic enzyme responsible for changes in the metabolic system of HSCs under stress has not been identified. HSCs are essential for cell therapy, including HSC transplantation, and in order to comprehensively elucidate the metabolic systems that have attracted attention as their regulatory mechanisms, recent studies have included metabolomic analyses using rare cell types such as HSCs^{(23), (46)–(49)}, as well as isotope tracer analyses of undifferentiated hematopoietic cells purified after *in vivo* administration of isotopic glucose⁽⁵⁰⁾. Although these approaches are useful for obtaining comprehensive information on intracellular metabolites, they are not suited to track real-time changes in cellular metabolism at high resolution. Therefore, new approaches are necessary to analyze metabolites quantitatively and continuously without disturbing the physiological states of single cells while integrating the recently reported metabolome analysis techniques. In this study, we aimed to identify the key metabolic enzymes that define metabolic differences between steady-state and stress conditions in HSCs and elucidate their regulatory mechanisms using a quantitative and mathematical approach. Our findings provide a platform for quantitative metabolic analysis of rare cells such as HSCs, characterize the overall metabolic reprogramming of HSCs during stress loading, and highlight the key enzyme involved in this process.

Methods

Mice

C57BL/6 mice (7–16 weeks old, Ly5.2⁺) were purchased from Japan SLC (Shizuoka, Japan). C57BL/6 mice (Ly5.1⁺) were purchased from CLEA Japan (Shizuoka, Japan). Knock-in mice harboring GO-A^{Team2}^{(51)–(53)} in the ROSA26 locus were generated in the Yamamoto laboratory. The GO-A^{Team2} mice (8–16 weeks old) were used to analyze HSPCs. mVenus-p27K[−] mice (17–20 weeks old) were provided by the Kitamura Laboratory at The Institute of Medical Science, The University of Tokyo and used for cell cycle analysis^{(54), (55)}. Mice were genotyped using PCR-based assays of tail DNA or transdermal enhanced green fluorescent protein (EGFP) fluorescence. All mice were maintained in the animal facility at the National Center for Global Health and Medicine Research Institute under specific pathogen-free conditions and fed *ad-libitum*. Mice were euthanized by cervical dislocation. All animal experiments were approved by the Institutional Animal Care and Use Committee (IACUC) at

National Center for Global Health and Medicine Research Institute. Both male and female mice were used in experiments.

Ion chromatography mass spectrometry (IC-MS) analysis

For metabolome analysis focused on glycolytic metabolites and nucleotides, anionic metabolites were measured using an orbitrap-type MS (Q-Exactive Focus; Thermo Fisher Scientific, Waltham, MA, USA) connected to a high-performance IC system (ICS-5000+, Thermo Fisher Scientific), enabling highly selective and sensitive metabolite quantification owing to the IC-separation and Fourier Transfer MS principle⁽⁵⁶⁾. The IC instrument was equipped with an anion electrolytic suppressor (Dionex AERS 500; Thermo Fisher Scientific) to convert the potassium hydroxide gradient into pure water before the sample entered the mass spectrometer. Separation was performed using a Dionex IonPac AS11-HC-4 μ m IC column (Thermo Fisher Scientific). The IC flow rate was 0.25 mL/min supplemented post-column with 0.18 mL/min makeup flow of MeOH. The potassium hydroxide gradient conditions for IC separation were as follows: 1–100 mM (0–40 min), 100 mM (40–50 min), and 1 mM (50.1–60 min), with a column temperature of 30 °C. The Q-Exactive Focus mass spectrometer was operated under ESI negative mode for all detections. Full mass scan (m/z 70–900) was performed at a resolution of 70,000. The automatic gain control target was set at $3 \times 10^{(6)}$ ions, and the maximum ion injection time was 100 ms. Source ionization parameters were optimized with spray voltage at 3 kV, and other parameters were as follows: transfer temperature, 320 °C; S-lens level, 50; heater temperature, 300 °C; sheath gas, 36; and aux gas, 10. Metabolite amounts were quantified from calibration curve data generated based on peak areas and respective metabolite amounts. Detailed methods for sample collection and IC-MS analysis are described in **SUPPLEMENTARY METHODS**.

Time-course analysis of ATP levels in single hematopoietic cells using GO-ATeam2 system

GO-ATeam2 is a ratiometric biosensor that monitors ATP concentration through Förster resonance energy transfer (FRET) from EGFP to the monomeric version of Kusabira Orange (mKO), regardless of the sensor expression levels⁽⁵²⁾. Surface-marker-stained bone marrow (BM) mononuclear cells (MNCs) from GO-ATeam2 mice were dispensed into basal medium (Ba-M) containing minimal salts, vitamins, and buffers (HEPES and sodium bicarbonate), but no glucose or mitochondrial substrates (Table S1), or into medium containing mitochondrial substrates, pyruvate, lactate, fatty acids, and amino acids, but no glucose (PLFA medium). The FRET/EGFP ratio was analyzed in a real-time with a BD FACSAria IIIu Cell Sorter (BD Biosciences, Franklin Lakes, NJ, USA) under ambient pressure, and ATP levels were calculated. Detailed methods for sample collection and ATP concentration calculation are provided in **SUPPLEMENTARY METHODS**.

Statistical analysis

Data are presented as means $\pm$ SD, unless otherwise stated. For multiple comparisons, statistical significance was determined by Tukey's multiple comparison test using the Tukey HSD function in R $\times$ 64 4.0.3 software (R Core Team, Vienna, Austria). A paired or unpaired two-tailed Student's *t*-test and two-way ANOVA with Sidak's test were used for experiments with two groups. A *p*-value <0.05 was considered statistically significant.

Data sharing

All relevant data are available from the corresponding author upon reasonable request.

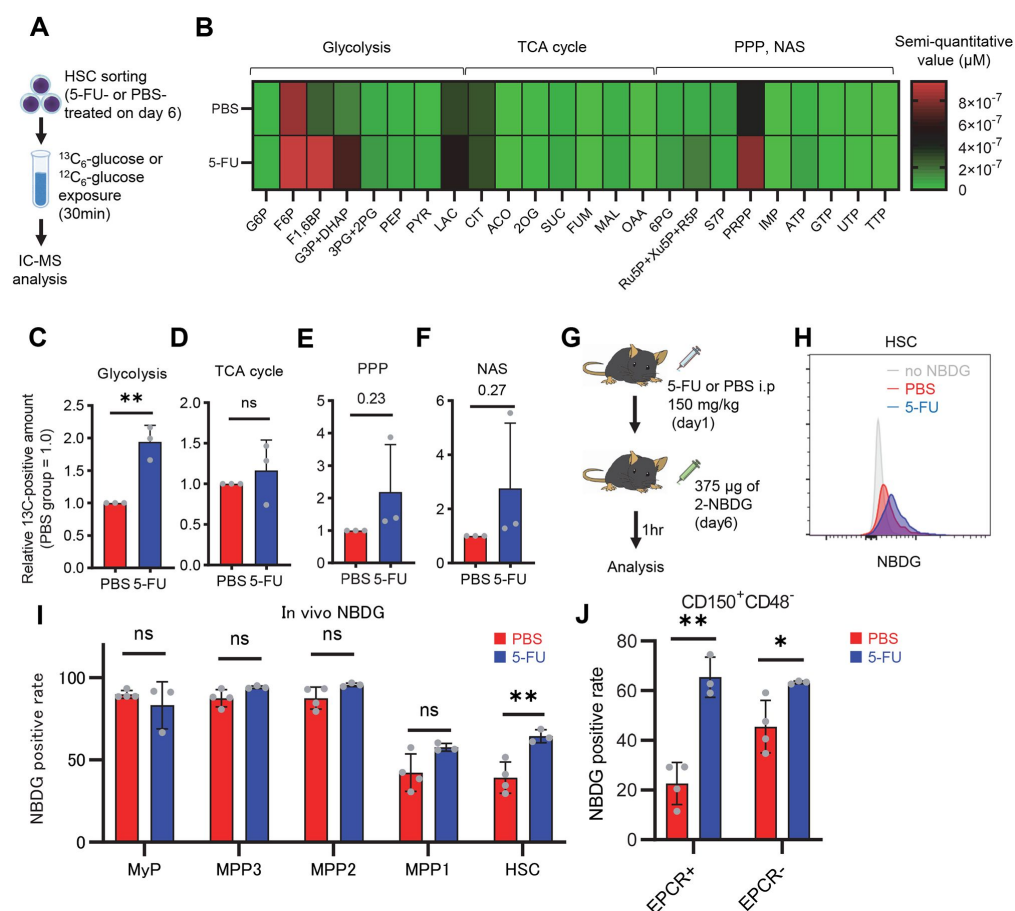
Results

HSC cell cycling increases anaerobic glycolytic flux

To determine how cell cycle progression alters HSC metabolism *in vivo*, we intraperitoneally and intravenously treated mice with 5-fluorouracil (5-FU) to induce HSC cell cycling (Supplemental Figure 1A). For analysis after 5-FU administration, the Lineage (Lin)[−] Sca-1⁺ c-Kit⁺ (LSK) gate was expanded to include as many HSCs as possible, for example high Sca-1-expressing cells and c-Kit-high to -dim Lin[−] cells, based on the previous report⁽⁵⁷⁾ (Supplemental Figure 1B). This expanded LSK gate was consistent with the patterns of Sca-1 and c-Kit expression observed in EPCR⁺ Lin[−] CD150⁺ CD48[−] cells, and the presence of EPCR, as a surface marker, identified HSCs with high stem cell activity after 5-FU administration⁽⁴¹⁾. We observed a transient decrease in the number of quiescent HSCs (Ki67[−]) and an increase in the number of cell-cycling HSCs (Ki67⁺) on day 6 after 5-FU treatment (Supplemental Figure 1C). Along with the loss of cell quiescence, ATP concentration in HSCs decreased transiently on day 6 (Supplemental Figure 1D). Because the route of administration of 5-FU (intraperitoneal or intravenous) made no difference in the Ki67 positivity rate of HSCs (Supplemental Figure 1E), we administered 5-FU intraperitoneally for remaining experiments. Two other methods were used to test whether cell cycle progression of HSCs after 5-FU treatment depends on the expression of EPCR. First, phosphorylation of Rb (pRb), a marker of cell cycle progression⁽⁵⁸⁾, was analyzed in HSCs after 5-FU treatment. Analysis of EPCR⁺ and EPCR[−] HSCs showed increased pRb in HSCs from 5-FU-treated mice in both fractions compared to HSCs from phosphate-buffered saline (PBS)-treated mice, regardless of EPCR expression (Supplemental Figure 1F-G). Second, we used a G₀ marker mouse line⁽⁵⁵⁾. These mice expressed a fusion protein of the p27 inactivation mutant p27K[−] and the fluorescent protein mVenus (G₀ marker), allowing prospective identification of G₀ cells. We tested whether the expression of G₀ marker in HSCs was altered after 5-FU administration to the G₀ marker mice (Supplemental Figure 1H) and found that 5-FU treatment reduced the frequency of G₀ marker-positive HSCs, regardless of the EPCR expression (Supplemental Figure 1I-J). This was not observed in the PBS group. These results indicated that 5-FU administration induced cell cycle progression of entire HSCs in mice.

HSC cell cycling is preceded by the activation of intracellular ATP-related pathways that metabolize extracellular nutrients, including glucose^{(39), (40)}, which are utilized in both ATP-producing and -consuming pathways, determining cellular ATP levels. Therefore, we examined the metabolic flux of glucose by performing *in vitro* IC-MS tracer analysis with uniformly carbon-labeled (U-¹³C₆) glucose to determine the pathways driving changes in ATP in 5-FU-treated HSCs (Figure 1A; Table S2). To avoid metabolite changes, samples were continuously chilled on ice during cell preparation, and the process from euthanasia to cell preparation was performed in the shortest possible time (see “**Preparation and storage of *in vitro* U-¹³C₆-glucose tracer samples**” section under “**Supplementary Methods**” for more information). We found that changes in metabolite levels before and after sorting were present but limited (Supplemental Figure 2A). This result is consistent with the finding that the cell purification process does not significantly affect metabolite levels when sufficient care is taken in cell preparation⁽⁵⁰⁾. In 5-FU-treated HSCs, the levels of glycolytic metabolites derived from U-¹³C₆-glucose were double those observed in PBS-treated HSCs (Figure 1B-C; Supplemental Figure 2B). The total levels of TCA cycle intermediates derived from U-¹³C₆-glucose were similar between PBS- and 5-FU-treated cells (Figure 1D; Supplemental Figure 2B). Levels of U-¹³C₆-glucose-derived intermediates involved in the pentose phosphate pathway (PPP) and nucleic acid synthesis (NAS) were two-fold higher in 5-FU-treated than in PBS-treated HSCs, whereas no significant differences in the levels of metabolites were observed between both groups (Figure 1E-F; Supplemental Figure 2B). Notably, the labeling

rate of metabolites during the first half of glycolysis was almost 100% in both groups, allowing us to easily track the labeled metabolites (Supplemental Figure 2C-E). This was thought to be due to the rapid replacement of unlabeled metabolites with labeled metabolites during exposure to $U\text{-}^{13}\text{C}_6\text{-glucose}$ because of the generally rapid glycolytic reaction. Next, we evaluated whether glucose uptake in HSCs after 5-FU administration was differentially affected by the expression of EPCR. The fluorescent analog of glucose, 2-(N-(7-nitrobenz-2-oxa-1,3-diazol-4-yl)amino)-2-deoxyglucose (2-NBDG), was administered intravenously to mice⁽⁵⁰⁾ and its uptake in EPCR⁺ and EPCR⁻ HSCs was assayed (Figure 1G). Regardless of the EPCR expression, the 2-NBDG uptake was greater in HSCs treated with 5-FU than in those treated with PBS (Figure 1H-J). Furthermore, compared with HSCs cultured under quiescence-maintaining conditions, those cultured under proliferative conditions were more resistant to OXPHOS inhibition by oligomycin (Supplemental Figure 1H; Table S1). Overall, the results showed that 5-FU-treated HSCs exhibited activated glycolytic flux, increasing the turnover of ATP. Moreover, glycolytic flux into mitochondria was equally unchanged in PBS- and 5-FU-treated-HSCs, supporting that 5-FU activated anaerobic glycolysis in HSCs.



pooled, analyzed by IC-MS, quantified based on calibration curve data for each metabolite (see “**Ion chromatography mass spectrometry (IC-MS) analysis**” section in “**Methods**” for details). **(G)** Experimental schema of *in vivo* 2-NBDG analysis. **(H)** Representative histograms of 2-NBDG analysis (gray: no 2-NBDG, red: PBS group, blue: 5-FU group). **(I)** 2-NBDG positivity in each fraction; data represent four pooled biological replicates for the PBS group and three for the 5-FU group; MyP: myeloid progenitor. **(J)** EPCR expression and 2-NBDG positivity within HSC fractions. Data were extracted from each individual in (I).

Data are presented as mean $\pm$ SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by paired-samples *t*-test (C-F) and Student's *t*-test (I and J). Abbreviations: G6P, glucose-6-phosphate; F6P, fructose-6-phosphate; F1,6BP, fructose-1,6-bisphosphate; G3P, glycerol-3-phosphate; DHAP, dihydroxyacetone phosphate; 3PG, 3-phosphoglycerate; 2PG, 2-phosphoglycerate; PEP, phosphoenolpyruvate; PYR, pyruvate; LAC, lactate; Ac-CoA, acetyl-CoA; CIT, citrate; ACO, cis-aconitic acid, isocitrate; 2OG, 2-oxoglutarate; SUC, succinate; FUM, fumarate; MAL, malate; OAA, oxaloacetate; 6PG, glucose-6-phosphate; Ru5P, ribulose-5-phosphate; Xu5P, xylulose-5-phosphate; R5P, ribose-5-phosphate; S7P, sedoheptulose-7-phosphate; E4P, erythrose-4-phosphate; PRPP, phosphoribosyl pyrophosphate; IMP, inosine monophosphate; ATP, adenosine triphosphate; GTP, guanine triphosphate; UMP, uridine monophosphate; UTP, uridine triphosphate; TTP, thymidine triphosphate. See also Fig. S1 and S2.

OXPHOS-inhibited HSCs exhibit compensatory glycolytic flux

Previous studies using mouse models of mitochondrial disease or defects in genes involved in electron transport chain and OXPHOS suggest that mitochondrial energy production is essential for maintaining HSC function^{(21), (35)–(37)}, as is the glycolytic system. However, there have been no quantitative reports on how OXPHOS-inhibited HSCs can adapt their metabolism. To understand HSC metabolism under OXPHOS inhibition, we performed *in vitro* U-¹³C₆-glucose tracer analysis of oligomycin-treated HSCs (Figure 2A; Table S3). Similar to 5-FU-treated HSCs (Figure 1), oligomycin-treated HSCs exhibited glycolytic system activation (Figure 2B–C; Supplemental Figure 2B). Metabolite flux to the TCA cycle and PPP was unchanged, but flux to the NAS was significantly reduced in oligomycin-treated HSCs compared to that in steady-state HSCs (Figure 2D–F; Supplemental Figure 2B). The results suggested that OXPHOS-inhibited HSCs activated compensatory glycolytic flux and suppressed NAS flux. As with 5-FU-treated HSCs, analysis of oligomycin-treated HSCs also showed almost 100% labeling of metabolites in the first half of glycolysis (Supplemental Figure 2F–H), allowing us to easily track the labeled metabolites. To further validate the compensatory glycolytic activation of HSCs under OXPHOS inhibition, a Mito Stress test was performed on HSCs and other differentiated myeloid progenitors (MyPs, Lin[−]Sca-1[−]c-Kit⁺ (LKS[−]) cells). The results showed that extracellular acidification rates (ECAR) were elevated in HSCs after oligomycin treatment compared to before oligomycin treatment (Figure 2G–H). No increase in ECAR was observed in MyPs (Figure 2G–H), supporting that inhibition of OXPHOS activated anaerobic glycolysis specifically in HSCs.

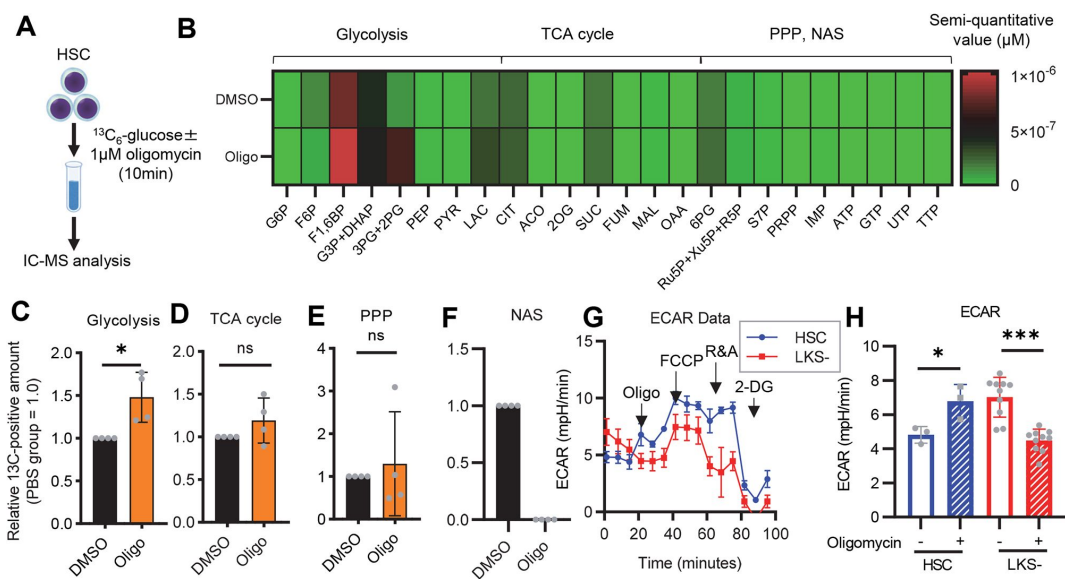


Figure 2.

OXPHOS inhibition activates compensatory glycolysis in HSCs.

(A) Experimental design used for glucose isotope tracer analysis in HSCs treated with the OXPHOS inhibitor oligomycin. **(B)** Heat map of metabolite levels detected by *in vitro* tracer analysis of U- $^{13}\text{C}_6$ -glucose in HSCs treated with DMSO or oligomycin (Oligo). **(C-F)** Relative amounts of U- $^{13}\text{C}_6$ -glucose-derived metabolites in glycolysis (C), the first round of TCA cycle (D), the P ϵ (E), and nucleotide synthesis (F) in DMSO-(black) or oligomycin-treated (orange) HSCs; In (B)-(F), biological replicates of the DMSO and oligomycin groups obtained on four separate days were pooled, analyzed by IC-MS, and quantified based on calibration curve data for each metabolite (see “**Ion chromatography mass spectrometry (IC-MS) analysis**” section in “**Methods**” for details). (G-H) Overview of the Mito Stress test on the Seahorse flux analyzer for HSC and MyPs; (G) ECAR before and after oligomycin treatment (H). (Data were obtained from $n = 3$ technical replicates for HSCs and $n = 10$ technical replicates for MyPs.)

Data are shown as mean $\pm$ SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by paired-samples *t*-test (C-E) and Student’s *t*-test (H). Abbreviations: G6P, glucose-6-phosphate; F6P, fructose-6-phosphate; F1,6BP, fructose-1,6-bisphosphate; G3P, glycerol-3-phosphate; DHAP, dihydroxyacetone phosphate; 3PG, 3-phosphoglycerate; 2PG, 2-phosphoglycerate; PEP, phosphoenolpyruvate; PYR, pyruvate; LAC, lactate; Ac-CoA, acetyl-CoA; CIT, citrate; ACO, cis-aconitic acid, isocitrate; 2OG, 2-oxoglutarate; SUC, succinate; FUM, fumarate; MAL, malate; OAA, oxaloacetate; 6PG, glucose-6-phosphate; Ru5P, ribulose-5-phosphate; Xu5P, xylulose-5-phosphate; R5P, ribose-5-phosphate; S7P, sedoheptulose-7-phosphate; E4P, erythrose-4-phosphate; PRPP, phosphoribosyl pyrophosphate; IMP, inosine monophosphate; ATP, adenosine triphosphate; GTP, guanine triphosphate; UMP, uridine monophosphate; UTP, uridine triphosphate; TTP, thymidine triphosphate. See also Fig. S2.

Phosphofructokinase (PFK) metabolism in HSCs is activated during proliferation and OXPHOS inhibition

To investigate whether glycolytic activation in HSCs after 5-FU treatment and OXPHOS inhibition could be demonstrated through unbiased mathematical simulations, we performed quantitative ^{13}C metabolic flux analysis (^{13}C -MFA). After generating a metabolic model for isotope labeling enrichment and setting appropriate lactate efflux values, a simulation was conducted using the labeled metabolite abundance data obtained from isotope tracer analysis. The appropriate lactate efflux for quiescent HSC (PBS-treated HSC) was determined to 65 after experimenting with values from 0–100. The lactate efflux of 5-FU- or oligomycin-treated HSCs was higher than that of quiescent HSCs based on the observation that labeled glycolytic metabolite levels were particularly elevated in *in vitro* tracer analysis (see “Quantitative ^{13}C -MFA with OpenMebius” under “Supplementary Methods” for more information). As a result, the variation in the flux values of all enzymatic reactions calculated in HSCs after 5-FU or oligomycin treatment became smaller compared to quiescent HSCs, suggesting that HSCs strictly regulated their metabolism in response to stress (Supplemental Figure 3A-C). Unlike PBS-treated HSCs, those treated with 5-FU or oligomycin exhibited preferential glycolytic activation rather than TCA- or PPP-based metabolic strategies; the first half of the glycolytic system appeared to be the site of metabolic activation (Figure 3A-D; Supplemental Figure 3D-U, Table S4). This increase in metabolic flux upstream of the glycolytic pathway was also supported by our *in vitro* tracer analysis (Figure 1B and Figure 2B), suggesting that ^{13}C -MFA was a valid metabolic simulation. Among the reactions in the first half of glycolysis, phosphorylation of fructose 6-phosphate by PFK is the irreversible and rate-limiting reaction⁽⁵⁹⁾. A detailed review of *in vitro* isotope tracer analysis results showed that the ratio of fructose 1,6-bisphosphate (F1,6BP; the product of PFK) to F6P (the substrate of PFK) was greatly elevated in HSCs during proliferation and OXPHOS inhibition (Figure 3J-K). Together with the results of quantitative ^{13}C -MFA, these findings suggested that HSCs exhibit elevated glycolytic flux relative to mitochondrial activity by increasing PFK enzyme activity under various stress conditions.

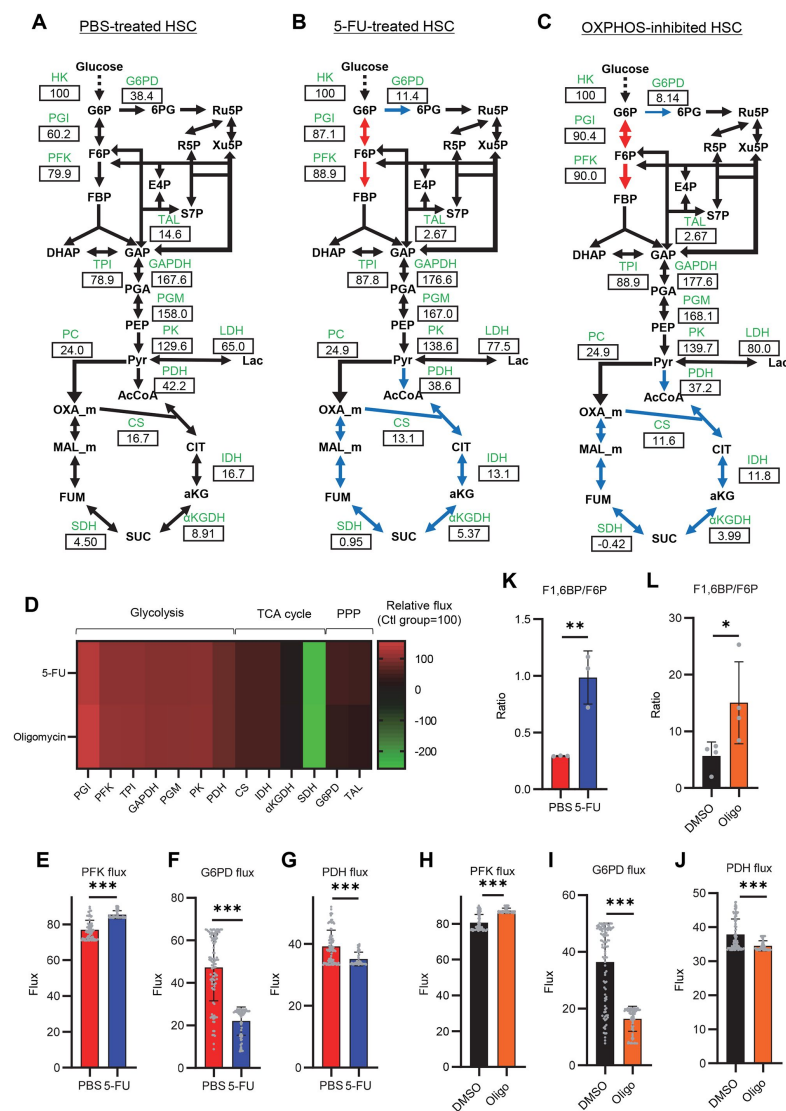


Figure 3.

Quantitative ^{13}C -MFA of quiescent, proliferative, and stressed HSCs

(A-C) Overview of quantitative ^{13}C -MFA of PBS-treated HSCs (A), 5-FU-treated HSCs (B), and OXPPOS-inhibited HSCs (C). The representative net flux for each reaction with glucose uptake as 100 is shown in the squares below the catalytic enzymes for each reaction listed in green letters. Red arrows indicate reactions with particularly elevated fluxes and blue arrows indicate reactions with particularly decreased fluxes. (D) Heatmap of the relative flux of each enzyme in the 5-FU or oligomycin groups compared to that in the quiescent (Ctl) HSC (The metabolic flux of each enzyme in the Ctl group was standardized as 100.). (E-J) Fluxes due to reactions with PFK (E, H), G6PD (F, I), and PDH (G, J). Fluxes of HSCs derived from mice treated with 5-FU (blue bars) or PBS (red bars) (D-F) and of HSCs treated with DMSO (black bars) or oligomycin (orange bars) (G-I) are shown. Data is obtained from 100 simulations in OpenMebius, and flux data for each enzyme is displayed (Table S4). (K-L) Ratio of fructose 1,6-bisphosphate (F1,6BP) to fructose-6-phosphate (F6P) calculated from tracer experiments shown in Figure 1B and Figure 2B. Effects of 5-FU administration (K) or mitochondrial inhibition by oligomycin (L) are summarized. Data are shown as mean $\pm$ SD. * $p \leq 0.05$, ** $p \leq 0.01$,

*** $p \leq 0.001$ as determined by Student's *t*-test (E-L). Abbreviations: HK, hexokinase; PGI, glucose-6-phosphate isomerase; PFK, phosphofructokinase; TPI, triose phosphate isomerase; GAPDH, glyceraldehyde-3-phosphate dehydrogenase; PGM, phosphoglycerate mutase; PK, pyruvate kinase; LDH, lactate dehydrogenase; PC, pyruvate carboxylase; PDH, pyruvate dehydrogenase; CS, citrate synthase; IDH, isocitrate dehydrogenase; α KGDH, α -ketoglutaric acid dehydrogenase; SDH, succinate dehydrogenase; G6PD, glucose-6-phosphate dehydrogenase; TAL, transaldolase. See also Fig. S3.

HSCs under stress exhibit activation of glycolysis-initiated TCA cycle and NAS

To investigate the long-term glucose utilization of HSCs, we performed an *in vivo* tracer analysis with $\text{U-}^{13}\text{C}_6$ glucose based on recent reports^{(47), (50)} (Supplemental Figure 4A; see “Preparation and storage of *in vivo* $\text{U-}^{13}\text{C}_6$ -glucose tracer samples” under “Supplementary Methods” for more information). In HSCs from 5-FU-treated mice, we observed increased labeling of glycolytic metabolites such as dihydroxyacetone phosphate, glycerol-3-phosphate, and phosphoenolpyruvate, as well as NAS metabolites such as inosine

monophosphate and ATP, and those derived from TCA cycle such as aspartic acid and glutamate, compared to HSCs from PBS-treated mice (Supplemental Figure 4B-I, Table S5). When the amount of U- $^{13}\text{C}_6$ -glucose-derived labeled metabolites in each pathway was calculated, more glucose-derived metabolites entered TCA cycle in the 5-FU-treated group than PBS-treated group (Supplemental Figure 4J). Thus, although short-term (10–30 min) *in vitro* tracer analysis showed that HSCs exhibited more potent activation of anaerobic glycolysis than of other pathways in response to 5-FU administration, long-term (approx. 3 h) labeling by *in vivo* tracer analysis revealed that glycolysis-initiated TCA cycle and NAS flux were activated in addition to enhanced anaerobic glycolysis.

PFKFB3 accelerates glycolytic ATP production during HSC cell cycling

In vitro and *in vivo* tracer analysis results collectively suggested that the activation of glycolysis catalyzed by PFK may have been the starting point for the activation of the entire HSC metabolism. To analyze the contribution of PFK to ATP metabolism in steady-state or stressed HSCs, we needed to develop an experimental system that could measure the dynamics of ATP concentrations in HSCs in a non-destructive, real-time manner. To this end, we used knock-in GO-ATeam2 mice as a FRET-based biosensor of ATP concentration (see “**Time-course analysis of ATP levels in single hematopoietic cells using GO-ATeam2 system**” under “**Methods**” for more information.). The number of BMMNCs, as well as the frequency of HSCs (CD150⁺CD48-LSK) and other progenitor cells, in the BM of GO-ATeam2⁺ mice were almost unchanged compared to of C57B6/J mice, except for a mild decrease in the frequency of the Lin[−] fraction (Supplementary Figure 5A-C). Using BMMNCs derived from GO-ATeam2⁺ mice, we developed a method to detect changes in ATP concentration with high temporal resolution when the activity of PFK was modulated (**METHODS**, Supplementary Figure 5D-F). To validate our methods, we measured ATP concentrations in HSCs and MyPs with or without various nutrients (see “**Time-course analysis of FRET values**” under “**Supplementary Methods**” for more information.). MyPs showed more rapid decreases in ATP concentration than HSCs, suggesting higher ATP consumption by progenitors (Supplementary Figure 5G-H). Adding glucose to the medium suppressed this decrease in MyPs; however, other metabolites (e.g., pyruvate, lactate, and fatty acids) had minimal effects, suggesting that ATP levels are glycolysis-dependent in MyPs (Supplementary Figure 5G-H), consistent with previous reports that the aerobic glycolytic enzyme M2 pyruvate kinase isoform (PKM2) is required for progenitor cell function⁽³³⁾.

Further, we analyzed ATP consumption and metabolic dependency of cell-cycling HSCs after 5-FU administration (Figure 4A). After inhibiting glycolysis using 2-deoxy-D-glucose (2-DG) with other mitochondrial substrates, 5-FU-treated HSCs showed more rapid decreases in ATP concentration than PBS-treated HSCs (Figure 4B-C). In contrast, OXPHOS inhibition by oligomycin without glucose or mitochondrial substrates decreased the ATP concentration to a similar extent in both 5-FU- and PBS-treated HSCs, although 5-FU-treated HSCs showed earlier ATP exhaustion (Figure 4D-E). These data suggest that 5-FU-treated-HSCs upregulated ATP production via glycolysis, rather than relying on mitochondria. Apoptosis assay revealed a slight increase in early apoptotic cells (annexin V⁺ propidium iodide [PI][−]) after 2-DG treatment and a slight decrease in the number of viable cells (Annexin V[−] PI[−]) after oligomycin treatment, both to a very limited extent (approx. 5%) compared to the degree of ATP decrease, suggesting that the decrease in ATP after 2-DG or oligomycin treatment did not simply reflect cell death (Supplementary Figure 5I).

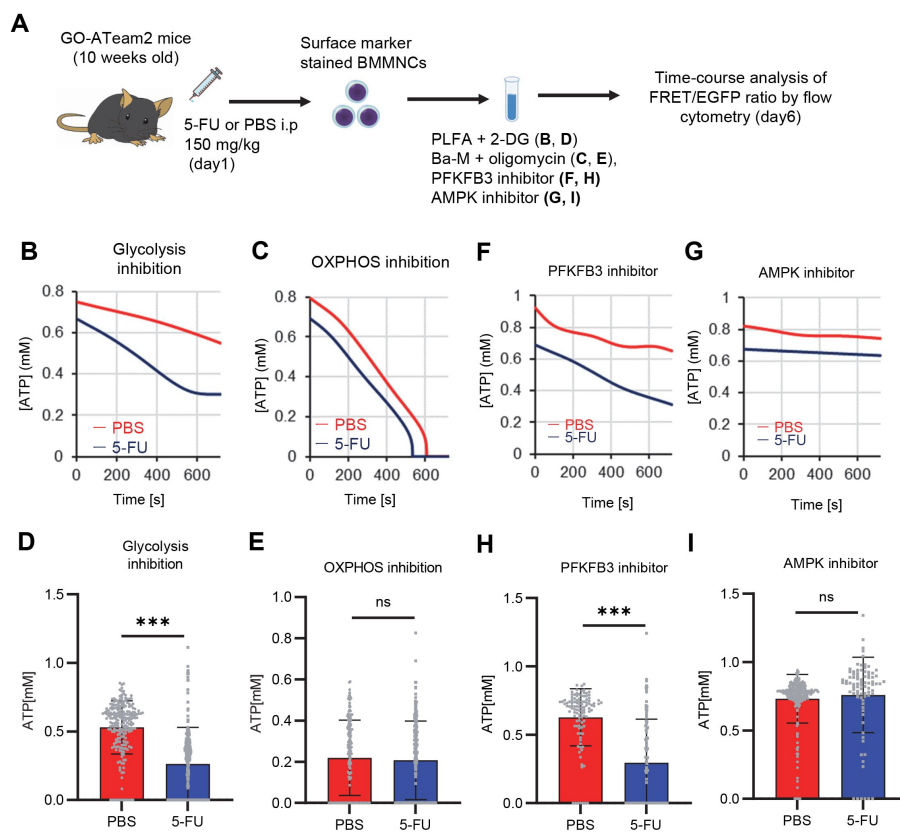


Figure 4.

Pfkfb3 activates the glycolytic system in proliferating HSCs

(A) Experimental design used to conduct real-time ATP analysis of HSCs treated with 5-FU or PBS. PLFA medium containing mitochondrial substrates (pyruvate, lactate, fatty acids, and amino acids) but no glucose, was used for experiments with 2-DG; Ba-M containing neither mitochondrial substrates nor glucose was used for experiments with oligomycin, Pfkfb3 inhibitor, or AMPK inhibitor.

(B–I) Results of real-time ATP analysis of PBS-(red) or 5-FU-treated (blue) HSCs after treatment with 2-DG (B, D), oligomycin (C, E), PFKFB3 inhibitor (F, H), or AMPK inhibitor (G, I). Bar graphs show corrected ATP concentrations for the last 2 min (D) of (B), 6–7 min (E) of (C), or the last 1

min (H, I) of (F, G) for PFKFB3 and AMPK inhibitors, respectively. Each group represents at least 60 cells. Data are representative results of pooled samples from three biological replicates. (See “Time-course analysis of FRET values” in “Supplementary Methods” for details of the correction method used to calculate ATP concentration.)

Data are presented as mean $\pm$ SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by Student’s *t*-test (D, E, H, and I). See also Fig. S5.

PFK is allosterically activated by 6-phosphofructo-2-kinase/fructose-2,6-bisphosphatase (PFKFB). Among the four isozymes of mammalian PFKFB, PFKFB3 is the most favorable for PFK activation⁽⁶⁰⁾. We investigated whether PFKFB3 contributes to glycolytic plasticity in HSCs during proliferation. When treated with the PFKFB3 inhibitor AZ PFKFB3 26, compared with HSCs from PBS-treated mice, HSCs from 5-FU-treated mice showed decreased ATP levels (Figure 4F, H, Supplementary Figure 5J). Although AMPK activates PFKFB3 in other contexts⁽⁶¹⁾, AMPK inhibition by dorsomorphin did not alter ATP concentration in 5-FU-treated-HSCs (Figure 4G, I).

OXPHOS inhibition accelerates glycolysis to sustain ATP levels in HSCs, but not in progenitors

To assess differences in metabolic dependence between steady-state or stressed and naturally proliferating HPCs, we altered ATP metabolism in HSCs and progenitors using 2-DG or oligomycin (Figure 5A). Oligomycin treatment rapidly depleted ATP in HSCs and all HPC fractions (green lines in Figure 5B–C; Supplemental Figure 6A–D). Treatment with 2-DG decreased ATP concentrations for a short amount of time (~12 min) in HSCs and HPCs, but

ATP reduction was less evident than that induced by oligomycin (blue lines in Figure 5B-C; Supplemental Figure 6A-D). The ATP reduction induced by 2-DG treatment was particularly low (~15%) in HSCs, multipotent progenitor cells (MPPs), and common lymphoid progenitors (CLPs) relative to that in common myeloid progenitors (CMPs), granulocytes-macrophage progenitors (GMPs), and megakaryocyte-erythrocyte progenitors (MEPs) (Figure 5D).

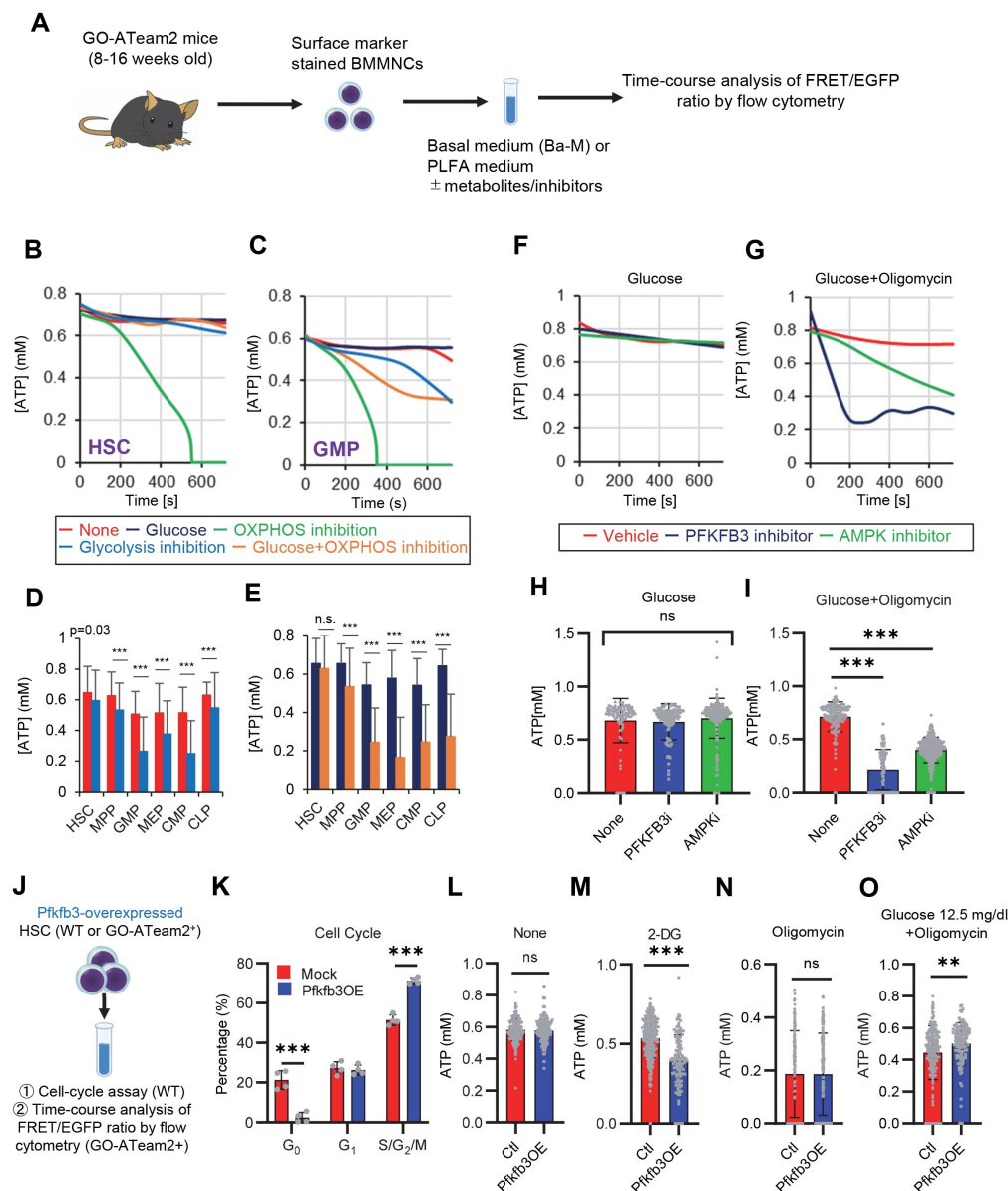


Figure 5.

PFKFB3 accelerates glycolysis in HSCs under OXPHOS inhibition in an AMPK-dependent manner

(A) Experimental design of real-time ATP analysis using GO-ATeam2 knock-in BMMNCs. Ba-M was used in experiments with oligomycin. For other experiments, PLFA medium was used. (B-C) Evaluation of factors affecting ATP concentration in HSCs (B) and GMPs (C) based on the GO-ATeam2 system. GO-ATeam2 knock-in BMMNCs were incubated with glucose, oligomycin, 2-DG, or glucose plus oligomycin, and the FRET/EGFP ratio was calculated. (D) ATP concentration in indicated stem/progenitor fractions in PLFA medium (red bars) alone or PLFA medium plus 2-DG (blue bars). ATP concentration for the last 2 min of the analysis time is shown. Data is summarized

from Figure 5B-C and Supplemental Figure 5A-D. Each group represents at least 110 cells. Data are representative results of pooled samples from three biological replicates. (E) ATP concentration in indicated stem/progenitor fractions in Ba-M plus glucose (dark blue bars) or Ba-M plus glucose and oligomycin (orange bars). ATP concentration for the last 1 min of the analysis period is shown. Data is summarized from Figure 5B-C and Supplemental Figure 5A-D. Each group represents at least 43 cells. Data are representative results of pooled samples from three biological replicates. (F-I) Effects of PFKFB3 or AMPK inhibitors (PFKFB3i or AMPKi, respectively) on ATP concentration in HSCs from GO-ATeam2 mice in Ba-M plus glucose only (F) or Ba-M plus glucose and oligomycin (G). ATP concentrations for the last 1 min of the analysis period are shown in (H) and (I) for glucose only and glucose with oligomycin groups, respectively. Each group represents at least 90 cells. Data are representative results of pooled samples from three biological replicates. (J) Experimental schema for cell

cycle assay and real-time ATP concentration analysis after overexpression of *Pfkfb3*. **(K)** Cell cycle status of *Pfkfb3*-overexpressing (*Pfkfb3*OE) and *mock*-transduced HSCs. **(L-O)** Effects of inhibitors on ATP concentration in *Pfkfb3*-overexpressing GO-ATeam2⁺ HSCs. Cells were exposed to vehicle (Ctl) (L), 2-DG (M), oligomycin (N), or glucose 12.5 mg/dL and oligomycin (O), and ATP concentrations for the last 1 (L, N, and O) or 2 min (M) of the analysis period were calculated. Data are representative results of pooled samples from three biological replicates.

Data are presented as mean $\pm$ SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by Student's *t*-test (D, E, and K-O) or one-way ANOVA followed by Tukey's test (H and I). See also Fig. S5.

Next, we investigated the role of glycolysis in ATP production during OXPHOS inhibition by combining oligomycin administration and glucose supplementation. ATP concentration remained more stable in HSCs treated with oligomycin and glucose than in those treated only with oligomycin. Similar results were not seen in HPCs, indicating that HSCs have the plasticity to upregulate glycolytic ATP production to meet demands (orange lines in [Figure 5B-C](#); Supplemental Figure 6A-D, summarized in [Figure 5E](#)). Similar to oligomycin treatment, rotenone (Complex I inhibitor) and carbonyl cyanide 4-(trifluoromethoxy)phenylhydrazone (FCCP, mitochondrial uncoupler) treatments, which inhibit OXPHOS-derived ATP production, also decreased ATP concentrations in HSCs, but not when administered simultaneously with glucose (Supplementary Figure 6E-F). Furthermore, with oligomycin, HSCs, but not HPCs, maintained ATP concentrations at low glucose levels (50 mg/dL) (Supplemental Figure 6G). These analyses suggest that ATP was produced by mitochondrial OXPHOS in steady-state HSCs, and that only HSCs, but not HPCs, maintained ATP production by glycolysis when OXPHOS was compromised.

PFKFB3 accelerates glycolytic ATP production during OXPHOS inhibition

Next, to understand whether PFKFB3 contributes to ATP production in HSCs under OXPHOS inhibition, we evaluated PFKFB3 function under OXPHOS inhibition using the GO-ATeam2⁺ BMMNCs. In oligomycin-treated HSCs, PFKFB3 inhibition led to rapidly decreased ATP concentration that was not observed in HSCs not treated with oligomycin ([Figure 5F-I](#)). We examined the effects of HSPC metabolic regulators on ATP levels in oligomycin-treated HSCs. Inhibiting PKM2, which accelerates glycolysis in steady-state progenitors⁽³³⁾, significantly reduced ATP levels in oligomycin-treated HSCs (Supplemental Figure 6H, J). Inhibiting LKB1, a kinase upstream of AMPK^{(62), (63)}, did not affect the ATP concentration in oligomycin-treated HSCs (Supplemental Figure 6I, K), whereas levels of adenosine monophosphate (AMP), which also activates AMPK, increased in oligomycin-treated but not in 5-FU-treated HSCs (Supplemental Figure 6L). This may explain differences in AMPK-dependent ATP production between proliferative HSCs and HSCs under OXPHOS inhibition.

Next, we tested the effects of PFKFB3 on ATP concentration in HPCs. Unlike HSCs, HPCs exhibited PFKFB3-dependent ATP production, even without oligomycin (Supplemental Figure 6M-Q). Therefore, ATP production in steady-state HSCs was PFKFB3-independent, and proliferative stimulation or OXPHOS inhibition plastically activated glycolytic ATP production in a PFKFB3-dependent manner to meet ATP demand.

PFKFB3 activity renders HSCs dependent on glycolysis

Next, we investigated whether PFKFB3 activity itself confers glycolytic dependence on HSCs. We retrovirally overexpressed *Pfkfb3* in HSCs and performed cell cycle analysis ([Figure 5J](#)). *Pfkfb3*-overexpressed HSCs increased the proportion of cells in the S/G2/M phase and

decreased the number of G_0 cells compared to *mock*-overexpressed HSCs (Figure 5K). Next, we retrovirally overexpressed *Pfkfb3* in GO-A^{Team2+} HSCs and performed real-time ATP measurement (Figure 5J). *Pfkfb3*-overexpressing GO-A^{Team2+} HSCs did not show changes in ATP concentrations relative to those in *mock*-transduced cells (Figure 5L; Supplemental Figure 6R). Upon 2-DG treatment, *Pfkfb3*-overexpressing HSCs showed a greater decrease in ATP concentration than *mock*-transduced HSCs did (Figure 5M; Supplemental Figure 6S). However, oligomycin treatment of both *mock*-transduced and *Pfkfb3*-overexpressing HSCs decreased ATP concentration to comparable levels (Figure 5N; Supplemental Figure 6T). Notably, *Pfkfb3*-overexpressing HSCs recovered ATP levels more effectively under low glucose conditions (12.5 mg/dL) than did *mock*-transduced HSCs (Figure 5O; Supplemental Figure 6U). These data suggest that PFKFB3 directly conferred glycolytic dependence onto HSCs by modulating the cell cycle and increasing their ATP-generating capacity via glycolysis under metabolic stress.

PFKFB3 methylation by PRMT1 supports ATP production by cell-cycling HSCs

Next, we investigated how 5-FU-treated-HSCs regulate PFKFB3 independently of AMPK (Figure 4G, I). PFKFB3 activity is regulated at multiple levels⁽⁶⁴⁾, and PFKFB3 transcript and protein levels in HSCs remained unchanged during 5-FU-induced cell cycling (Figure 6A-B). Phosphorylation can also regulate PFKFB3 activity^{(61), (65), (66)}; however, we observed no change in PFKFB3 phosphorylation in 5-FU-treated-HSCs (Figure 6C). Upon oligomycin exposure, PFKFB3 was phosphorylated by AMPK in the HSCs (Figure 6D). PFKFB3 is also methylated, and its activity is upregulated by protein arginine methyltransferase 1 (PRMT1)⁽⁶⁷⁾. We observed that *Prmt1* expression increased in 5-FU-treated-HSCs relative to that in PBS-treated-HSCs (Figure 6E). Furthermore, PFKFB3 methylation was significantly induced in 5-FU-treated-HSCs than in PBS-treated-HSCs (Figure 6F). Treatment of HSCs with a PRMT1 inhibitor decreased PFKFB3 methylation (Figure 6G), suggesting that PRMT1 catalyzed PFKFB3 methylation. To investigate whether glycolytic activity in HSCs was regulated by m-PFKFB3, mice treated with PBS or 5-FU were injected with 2-NBDG, and m-PFKFB3 levels in HSCs with high and low 2-NBDG uptake were quantified. Regardless of PBS or 5-FU treatment, HSCs with high 2-NBDG uptake exhibited higher m-PFKFB3 levels than those with low uptake (Figure 6H), suggesting that m-PFKFB3 regulated the activity of the glycolytic system in HSCs.

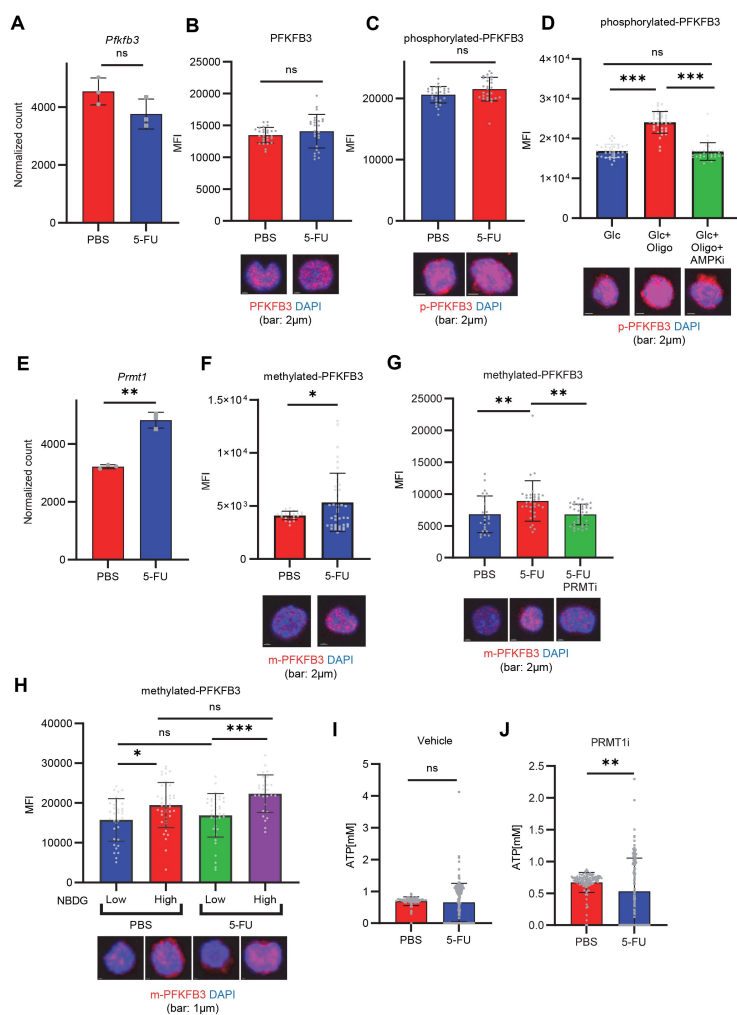


Figure 6.

PFKFB3 methylation by PRMT1 enables ATP production by cell-cycling HSCs.

(A) Normalized *Pfkfb3* mRNA counts based on RNA sequencing of PBS-treated (red) or 5-FU-treated (blue) HSCs. Data are representative results of pooled samples from three biological replicates. **(B)** Quantification of mean fluorescent intensity (MFI) of PFKFB3 protein in PBS- or 5-FU-treated HSCs. The lower part of the graph shows representative images of immunocytochemistry of PFKFB3 in each group. **(C)** Quantification of MFI of phosphorylated-PFKFB3 (p-PFKFB3) protein in PBS- or 5-FU-treated HSCs. The lower part of the graph shows representative images of immunocytochemistry of p-PFKFB3 in each group. **(D)** Quantification of mean fluorescence intensity (MFI) of p-PFKFB3 in HSCs treated with glucose (200mg/dL); glucose plus oligomycin (1 μ M); and glucose, oligomycin, and dorsomorphin (100 μ M) for 5 min. The lower part of the graph shows representative images of immunocytochemistry of p-PFKFB3 in each group. **(E)** Normalized *Prmt1* mRNA counts based on RNA sequencing of PBS-treated (red) or 5-FU-treated (blue) HSCs. Data are representative results of pooled samples from three biological replicates. **(F)** MFI quantification of methylated-PFKFB3 (m-PFKFB3) in PBS- or 5-FU-treated HSCs. The lower part of the graph shows representative images of immunocytochemistry of m-PFKFB3 in each group. **(G)** Quantification of MFI of m-PFKFB3 in PBS- or 5-FU-treated HSCs or 5-FU-treated HSCs after 15 min treatment with a PRMT1 inhibitor (90 μ g/mL GSK3368715); **(H)** Quantitation of m-PFKFB3 in NBDG-positive or -negative HSCs in mice treated with PBS or 5-FU. The lower part of the graph shows representative images of immunocytochemistry of m-PFKFB3 in each group. **(I-J)** Corrected ATP levels in PBS- (red) or 5-FU-treated (blue) HSCs 15 min after treatment with vehicle (I) or a PRMT1 inhibitor (90 μ g/mL GSK3368715) (J). Each group represents at least 101 cells. Data are representative results of pooled samples of two biological replicates. (see “Time-course analysis of FRET values” in “Supplementary Methods” for details of the correction method used to calculate ATP concentration.)

from two independent experiments. **(E)** Normalized *Prmt1* mRNA counts based on RNA sequencing of PBS-treated (red) or 5-FU-treated (blue) HSCs. Data are representative results of pooled samples from three biological replicates. **(F)** MFI quantification of methylated-PFKFB3 (m-PFKFB3) in PBS- or 5-FU-treated HSCs. The lower part of the graph shows representative images of immunocytochemistry of m-PFKFB3 in each group. **(G)** Quantification of MFI of m-PFKFB3 in PBS- or 5-FU-treated HSCs or 5-FU-treated HSCs after 15 min treatment with a PRMT1 inhibitor (90 μ g/mL GSK3368715); **(H)** Quantitation of m-PFKFB3 in NBDG-positive or -negative HSCs in mice treated with PBS or 5-FU. The lower part of the graph shows representative images of immunocytochemistry of m-PFKFB3 in each group. **(I-J)** Corrected ATP levels in PBS- (red) or 5-FU-treated (blue) HSCs 15 min after treatment with vehicle (I) or a PRMT1 inhibitor (90 μ g/mL GSK3368715) (J). Each group represents at least 101 cells. Data are representative results of pooled samples of two biological replicates. (see “Time-course analysis of FRET values” in “Supplementary Methods” for details of the correction method used to calculate ATP concentration.)

Data are presented as mean $\pm$ SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by Student's *t*-test (A-C, E-F, and I-J) or one-way ANOVA followed by Tukey's test (D, G, and H).

Further, we analyzed the potential effects of PRMT1 inhibition on ATP concentration in GO-ATeam2⁺ HSCs. Treatment with the PRMT1 inhibitor significantly decreased ATP levels in 5-FU-treated-HSCs than in PBS-treated-HSCs (Figure 6I-J). These findings indicated that ATP levels in 5-FU-treated-HSCs were supported by PRMT1 methylation-mediated PFKFB3 activation.

PFKFB3 contributes to HSPC pool expansion and stress hematopoiesis maintenance

Finally, we analyzed PFKFB3 function in HSCs during hematopoiesis. We cultured HSCs with a PFKFB3 inhibitor *in vitro* under quiescence-maintaining or proliferative conditions (Supplementary Figure 7A)⁽⁶⁸⁾. Cell count in HSC-derived colonies decreased following treatment with a PFKFB3 inhibitor under proliferative, but not quiescence-maintaining, conditions (Supplementary Figure 7B).

We also knocked out *Pfkfb3* in HSCs using the less toxic, vector-free CRISPR-Cas9 system and cultured the cells under quiescence-maintaining or proliferative conditions (Supplementary Figure 7A) based on recent reports by Shiroshita et al.⁽⁶⁹⁾. Again, cell numbers in *Pfkfb3*-knockout (KO) HSC-derived colonies decreased only in proliferative cultures when compared to control cultures (*Rosa26*-KO HSCs) (Supplementary Figure 7C, E, F). We retrovirally overexpressed *Pfkfb3* in HSCs and cultured them under quiescence maintenance or proliferative conditions (Supplementary Figure 7A). *Pfkfb3*-overexpressing HSC colonies showed increased cell count compared to that of *mock*-transduced cells, but only under proliferative conditions (Supplementary Figure 7D).

To assess PFKFB3 function in HSCs *in vivo*, we transplanted *Pfkfb3*-KO HSCs (Ly5.2⁺) or wild type (WT) control HSCs into lethally irradiated recipients (Ly5.1⁺) as well as Ly5.1⁺ competitor cells (Figure 7A), and the behavior of *Pfkfb3*-KO cells was evaluated by Sanger sequencing of peripheral blood (PB) cells⁽⁶⁹⁾. In the KO group, donor-derived chimerism in PB cells decreased relative to that in the WT control group during the early phase (1 month post-transplant) but recovered thereafter (Figure 7B). Next, we retrovirally transduced Ly5.2⁺ HSCs with *Pfkfb3* S461E (*Pfkfb3*CA), a constitutively active PFKFB3 mutant, and transplanted them into lethally irradiated recipients (Ly5.2⁺), along with Ly5.1⁺ competitor cells (Figure 7A, Supplementary Figure 7G). Donor chimerism during the early post-transplant period in the *Pfkfb3*CA-overexpressing group was significantly higher than that in the *mock*-transduced group (Figure 7C). These findings suggested that PFKFB3 was required for differentiation and proliferation of HSCs to expand the HSPC pool. Therefore, we compared the contribution of PFKFB3 to HSPC function at steady state and after myeloproliferative stimulation. *Pfkfb3*- or *Rosa*-KO HSPCs were transplanted into recipients (Ly5.1⁺). After 2 months, recipients received 5-FU intraperitoneally, and the dynamics of *Pfkfb3*- or *Rosa*-KO cell abundance in PB was assessed (Figure 7D). In PB cells prior to 5-FU administration, *Pfkfb3*- or *Rosa*-KO HSPC-derived blood cells were almost equally present, suggesting that the loss of PFKFB3 did not affect steady-state blood cell production (Figure 7E). However, after 5-FU administration, *Pfkfb3*-KO HSPC-derived blood cell abundance was reduced compared to that in the *Rosa* group (Figure 7E). This change occurred on day 6 after 5-FU administration (day 1), when the cell cycle of HSCs was activated (Supplementary Figure 1B), suggesting that PFKFB3 served a function in the proliferation and differentiation of HSPCs from HSCs.

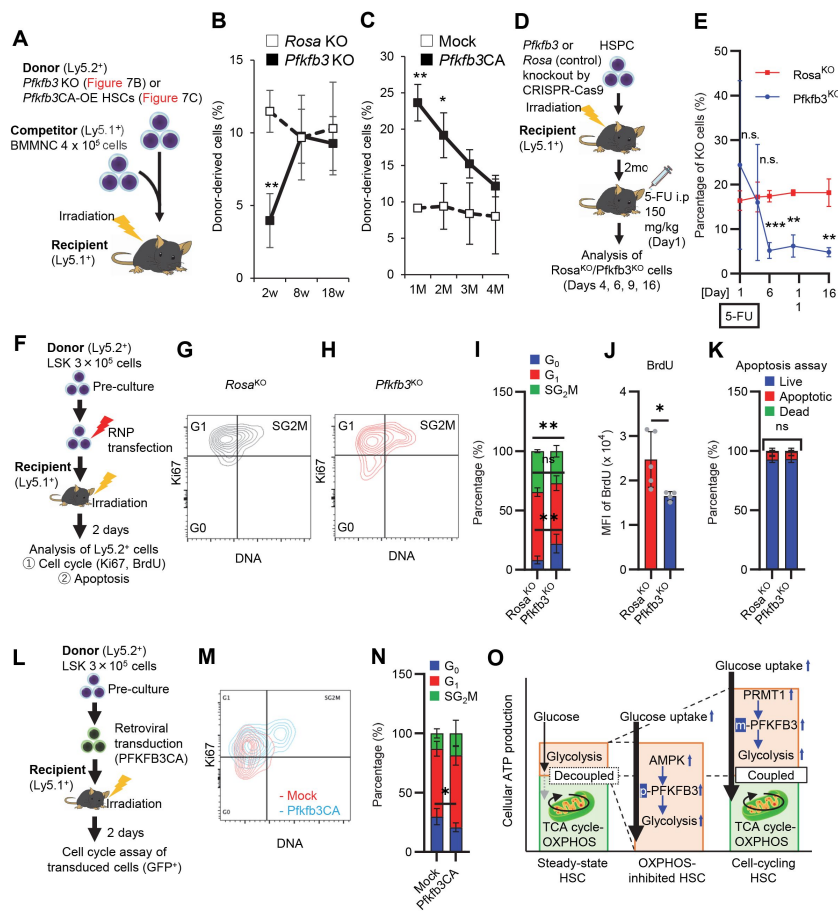


Figure 7.

PFKFB3 maintains HSC function under proliferative stress.

(A–C) Transplant analysis of *Pfkfb3*-KO or *Pfkfb3*CA-overexpressing HSCs. Experimental design (A). PB chimerism of donor-derived cells at 4 months post-transplant. *Pfkfb3*-KO group, n = 6; *Rosa*-KO group, n = 4; (B) *Pfkfb3* group, n = 5; pMY-IRES-GFP group, n = 4. (C) The data are representative results from two independent experiments. (D–E) 5-FU administration after bone marrow reconstruction with *Pfkfb3*- or *Rosa*-KO HSPCs. Experimental schema (D). Behavior of the *Pfkfb3*- or *Rosa*-KO cells in PB after 5-FU administration (E). n = 5 for each group. (F–K) Cell cycle analysis and apoptosis assay of *Pfkfb3*- or *Rosa*-KO HSPCs on day 2 post-BMT. Experimental schema (F). Representative plots of Ki67/Hoechst33432 staining of *Rosa*-KO (G) or *Pfkfb3*-KO (H) HSPCs and summary of analysis (I); summary of *in vivo* BrdU labeling assay (J). Apoptosis assay results (K). n = 4–5 biological replicates for each group. (L–N) Cell cycle analysis of *Pfkfb3*CA or Mock-overexpressing HSPCs on day 2 after BMT. Experimental Schema (L). Representative plot of Ki67/Hoechst33432 staining for both groups (M) and summary of analysis (N). n = 5 biological replicates for each group. (O) Models showing ATP production and regulation in quiescent, OXPHOS-inhibited, and cell-cycling HSCs. Note that the GO-ATeam2 system identified plastic acceleration of glycolysis by PFKFB3 in response to different types of stress maintains ATP levels.

Data are presented as mean ± SD. * p ≤ 0.05, ** p ≤ 0.01, *** p ≤ 0.001 as determined by Student's *t*-test (B, C, E, H, J, K, and N).

To investigate the mechanisms underlying the short-term effects of PFKFB3 on hematopoiesis after bone marrow transplantation (BMT), we evaluated cell cycle and apoptosis of *Pfkfb3*-KO or -overexpressing HSPCs on day 2 after BMT (Figure 7F). Cell cycle was analyzed by Ki67/Hoechst33432 staining and *in vivo* BrdU labeling⁽⁵⁰⁾, which showed that cell cycle progression was suppressed in *Pfkfb3*-KO HSPCs (Figure 7G–J). In contrast, *Pfkfb3* KO cells did not exhibit aberrant apoptosis of HSPCs after BMT (Figure 7K). Furthermore, we examined the cell cycle of HSPCs overexpressing *Pfkfb3*CA on day 2 after BMT (Figure 7L) and found that *Pfkfb3*CA-overexpressing HSPCs showed accelerated cell cycle compared to mock-overexpressing HSPCs (Figure 7M–N). These data indicate that PFKFB3 governs cell cycle progression and promotes the production of differentiated cells from HSCs in proliferative environments, both *in vitro* in the presence of cytokine stimulation and immediately after transplantation *in vivo*.

Discussion

In this study, by combining metabolomic tracing of U- $^{13}\text{C}_6$ -labeled glucose and ^{13}C -MFA, we quantitatively identified the metabolic programs used by HSCs during steady-state, cell-cycling, and OXPHOS inhibition. Under proliferative stress, HSCs uniformly shift from mitochondrial respiration to glycolytic ATP production and PPP activation, which represent hallmarks of cell-cycling mammalian cells⁽⁷⁰⁾. Previous reports have emphasized the importance of glycolysis in maintaining HSC quiescence, but have primarily analyzed HSCs in transplant assays, wherein HSCs must enter the cell cycle^{(32), (71)}. Prior analysis of repopulation capacity, which is positively correlated with enhanced glycolysis, may have overestimated glycolytic ATP production and overlooked mitochondrial ATP production during native hematopoiesis. In fact, some studies have suggested that OXPHOS activity is important for HSC maintenance and function⁽²¹⁾.

Our method was based on recently reported quantitative metabolic analysis techniques for very small numbers of cells^{(23), (46)–(50)}, such as HSCs, and expands our knowledge of HSC metabolism during stress hematopoiesis. In our study, 5-FU administration in mice transiently decreased ATP concentration in HSCs in parallel with cell cycle progression, suggesting that HSC differentiation and cell cycle progression are closely related to intracellular metabolism and can be monitored by measuring ATP concentration. We mainly analyzed a mixture of EPCR⁺ and EPCR[−] HSCs, and we believe that the observed cell cycle progression and promotion of glycolysis in both EPCR⁺ and EPCR[−] HSCs support the validity of our claims (Figure 1J, Supplementary Figure 1F–J). According to ^{13}C -MFA enzymatic reaction flux of PFK in 5-FU-treated HSCs indicated a relative increase of approximately 10%. However, the flux value obtained by ^{13}C -MFA was calculated with glucose uptake as 100. Thus, when combined with the overall increase in the glycolytic pool demonstrated by *in vitro* isotopic glucose tracer analysis and *in vivo* NBDG analysis, rapid acceleration of glycolysis becomes evident throughout the HSCs, including subpopulations that were less responsive to stress^{(43)–(45)}. These findings are consistent with reports suggesting that HSCs have relatively low biosynthetic activity^{(72), (73)} that is rapidly activated in response to cell proliferation stimuli^{(40), (74)}. Notably, we found that HSCs could accelerate glycolytic ATP production to fully compensate for mitochondrial ATP production under OXPHOS inhibition, a phenomenon that is difficult to identify without real-time ATP analysis. On the other hand, we observed that ATP production in steady-state or cell-cycling HSCs and in naturally proliferating HPCs depended more on mitochondrial OXPHOS than on glycolysis; inhibiting glycolysis in steady-state HSCs resulted in only mild ATP decreases. This is in contrast with the rapid decrease in ATP levels caused by OXPHOS inhibition. Because fatty acid β -oxidation is necessary for HSC self-renewal and quiescence^{(39), (68)}, fatty acids may support mitochondrial ATP production independently of fluxes from glycolysis. Furthermore, although glycolysis and TCA cycle are decoupled in steady-state HSCs, in response to cell cycle progression, anaerobic glycolytic metabolism in HSCs is enhanced (Figure 1) and fluxes to TCA cycle and PPP from the glycolytic system are also promoted (Supplementary Figure 4). The difference in the degree of coupling between glycolysis and the TCA cycle between *in vitro* and *in vivo* tracer analysis may be explained by the difference in labeling time (10–30 min vs. 3 h). Altogether, these data provide direct evidence that glycolysis and TCA cycle become functionally uncoupled in quiescent HSCs^{(32), (38)}. Our findings are also consistent with previous reports of OXPHOS activation associated with HSC proliferation^{(32), (36), (75), (76)}. In other words, HSCs exhibit an increased proportion of anaerobic glycolysis-derived ATP by PFKFB3 upon proliferation and OXPHOS inhibition; furthermore, the glycolytic system is the starting point of metabolic activation and is indispensable for the overall enhancement of HSC metabolism (Figure 7H).

HPCs and leukemic cells accelerate glycolytic ATP production using PKM2 for differentiation and transformation, respectively⁽³³⁾; however, we demonstrated that glycolytic acceleration does not fully compensate for mitochondrial ATP production in HPCs. Mechanistically, PFKFB3 increased glycolytic activity in HSCs to maintain ATP concentrations during proliferation and OXPHOS inhibition. Under steady-state conditions, naturally proliferating HPCs rely on PFKFB3 for ATP production, whereas HSCs do not. Therefore, we can infer that quiescent HSCs at steady state can produce ATP via PFKFB3 activation in response to stress, enabling ATP generation. Furthermore, overexpression of *Pfkfb3* in HSCs increased glycolytic dependency, suggesting that PFKFB3 itself can modulate metabolic dependency in HSCs. Changes in glycolytic dependency in HSCs overexpressing *Pfkfb3* may seem small (0.06–0.13 mM) (Figure 5M, O). However, it is noteworthy that the rate of the reaction catalyzed by PFK varies greatly within a very narrow range of ATP concentrations, less than 1 mM. Webb et al. analyzed the factors controlling PFK activity and reported that the reaction rate of PFK varies by approximately 40% in the 0.3–1 mM ATP concentration range⁽⁷⁷⁾. The reason that differences in glycolytic dependence could be detected in cells overexpressing *Pfkfb3* may be that the ATP concentration at the time of analysis was approximately 0.5–0.6 mM, which is within the range where a small change in ATP concentration can dynamically alter PFK activity.

PFKFB3 supports hematopoiesis in contexts that require robust HSPC proliferation *in vitro* and *in vivo*. We showed that the positive or negative effect of *Pfkfb3* overexpression or KO on differentiated blood cell production is gradually lost after BMT. This is because HSPCs require PFKFB3 for cell cycle progression during stress hematopoiesis in the early phase after BMT (Figure 7F–J, and L–N). However, HSCs no longer require PFKFB3 for a certain period of time after BMT, probably because they regain a quiescent state. This is consistent with the fact that inhibition of PFKFB3 in quiescent HSCs does not reduce the ATP concentration (Figure 5F, H), suggesting that the activity of PFKFB3 is plastically modified. HSC metabolic plasticity is also illustrated by the mode of PFKFB3 activation, differing depending on stress type. During proliferative stress, PRMT1 methylates PFKFB3 in the HSCs to promote glycolytic ATP production, a modification that increases its activity⁽⁶⁷⁾. PRMT1 is required for stress hematopoiesis⁽⁷⁸⁾, but its downstream targets in HSCs remain unclear. Our results strongly suggest that PRMT1 targets PFKFB3 to stimulate glycolysis in HSCs. In contrast, under OXPHOS inhibition, PFKFB3 phosphorylation by AMPK is induced—another modification that also upregulates its activity. These two PFKFB3 protein modifications allow for flexible regulation of ATP production by glycolysis, even under simultaneous and different stresses. In fact, the constitutively active S461E PFKFB3 mutant, designed to mimic phosphorylation in response to OXPHOS inhibition, enhanced HSC reconstitution capacity after transplantation, suggesting that even if PFKFB3 is activated by one stress (in this case, proliferative), it has the activation capacity to respond to a different stress (i.e., mitochondrial). Therefore, the functions of phosphorylated and methylated forms of PFKFB3 are to some extent interchangeable, and either modification can be used to handle diverse stresses.

In summary, we found that HSCs exhibit a highly dynamic range of glycolytic flux. Our study highlights glycolysis as a pivotal source of energy production in stressed HSCs, and indicates that OXPHOS, although an important source of ATP, can be uncoupled from glycolysis in steady-state HSCs without compromising ATP levels. Because multiple PFKFB3 modifications safeguard HSCs against different stresses by accelerating glycolysis, interventions targeting these might effectively induce or manage stress hematopoiesis. This study provides a platform for comprehensive and quantitative real-time analysis of ATP concentration and its dynamics in HSPCs. Our approach allows for analysis of metabolic programs in rare cells and detection of various metabolic activities within a diverse cell population, making it applicable to the analysis of various tissue systems in normal and diseased states.

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Conflict of interest

The authors declare no competing financial interests.

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Reviewer #1 (Public Review):

Watanuki et al used metabolomic tracing strategies of U-13C6-labeled glucose and 13C-MFA to quantitatively identify the metabolic programs of HSCs during steady-state, cell-cycling, and OXPHOS inhibition. They found that 5-FU administration in mice increased anaerobic glycolytic flux and decreased ATP concentration in HSCs, suggesting that HSC differentiation and cell cycle progression are closely related to intracellular metabolism and can be monitored by measuring ATP concentration. Using the GO-ATeam2 system to analyze ATP levels in single hematopoietic cells, they found that PFKFB3 can accelerate glycolytic ATP production during HSC cell cycling by activating the rate-limiting enzyme PFK of glycolysis. Additionally, by using Pfkfb3 knockout or overexpressing strategies and conducting experiments with cytokine stimulation or transplantation stress, they found that PFKFB3 governs cell cycle progression and promotes the production of differentiated cells from HSCs in proliferative environments by activating glycolysis. Overall, in their study, Watanuki et al combined metabolomic tracing to quantitatively identify metabolic programs of HSCs and found that PFKFB3 confers glycolytic dependence onto HSCs to help coordinate their response to stress. Even so, several important questions need to be addressed as below:

1. Based on previous reports, the authors expanded the LSK gate to include as many HSCs as possible (Supplemental Figure 1B). However, while they showed the gating strategy on Day 6 after 5-FU treatment, results from other time-points should also be displayed to ensure the strict selection of time-points.

2. In Figure 1, the authors examined the metabolite changes on Day 6 after 5-FU treatment. However, it is important to consider whether there are any dynamic adjustments to metabolism during the early and late stages of 5-FU treatment in HSCs compared to PBS treatment, in order to coordinate cell homeostasis despite no significant changes in cell cycle progression at other time-points.
3. As is well known, ATP can be produced through various pathways, including glycolysis, the TCA cycle, the PPP, NAS, lipid metabolism, amino acid metabolism and so on. Therefore, it is important to investigate whether treatment with 5-FU or oligomycin affects these other metabolic pathways in HSCs.
4. In part 2, they showed that oligomycin treatment of HSCs exhibited activation of the glycolytic system, but what about the changes in ATP concentration under oligomycin treatment? Are other metabolic systems affected by oligomycin treatment?
5. In Figure 5M, it would be helpful to include a control group that was not treated with 2-DG. Additionally, if Figure 5L is used as the control, it is unclear why the level of ATP does not show significant downregulation after 2-DG treatment. Similarly, in Figure 5O, a control group with no glucose addition should be included.
6. In this study, their findings suggest that PFKFB3 is required for glycolysis of HSCs under stress, including transplantation. In Figure 7B, the results showed that donor-derived chimerism in PB cells decreased relative to that in the WT control group during the early phase (1 month post-transplant) but recovered thereafter. Although the transplantation cell number is equal in two groups of donor cells, it is unclear why the donor-derived cell count decreased in the 2-week post-transplantation period and recovered thereafter in the Pfkfb3 KO group. Therefore, they should provide an explanation for this. Additionally, they only detected the percentage of donor-derived cells in PB but not from BM, which makes it difficult to support the argument for increasing the HSPC pool.
7. In Figure 7E, they collected the BM reconstructed with Pfkfb3- or Rosa-KO HSPCs two months after transplantation, and then tested their resistance to 5-FU. However, the short duration of the reconstruction period makes it difficult to draw conclusions about the effects on steady-state blood cell production.
8. PFK is allosterically activated by PFKFB, and other members of the PFKFB family could also participate in the glycolytic program. Therefore, they should investigate their function in contributing to glycolytic plasticity in HSCs during proliferation. Additionally, they should also analyze the protein expression and modification levels of other members. Although PFKFB3 is the most favorable for PFK activation, the role of other members should also be explored in HSC cell cycling to provide sufficient reasoning for choosing PFKFB3.
9. In this study, the authors identified PRMT1 as the upstream regulator of PFKFB3 that is involved in the glycolysis activation of HSCs. However, PRMT1 is also known to participate in various transcriptional activations. Thus, it is important to determine whether PRMT1 affects glycolysis through transcriptional regulation or through its direct regulation of PFKFB3? Additionally, the authors should investigate whether PRMT1i inhibits ATP production in normal HSCs. Moreover, could we combine Figure 6I and 6J for analysis. Finally, the authors could conduct additional rescue experiments to demonstrate that the effect of PRMT1 inhibitors on ATP production can be rescued by overexpression of PFKFB3.

Reviewer #2 (Public Review):

In the manuscript Watanuki et al. want to define the metabolic profile of HSCs in stress/proliferative (myelosuppression with 5-FU), and mitochondrial inhibition and homeostatic conditions. Their conclusions are that during proliferation HSCs rely more on glycolysis (as other cell types) while HSCs in homeostatic conditions are mostly dependent on mitochondrial metabolism. Mitochondrial inhibition is used to demonstrate that blocking mitochondrial metabolism results in similar features of proliferative conditions.

The authors used state-of-the-art technologies that allow metabolic readout in a limited number of cells like rare HSCs. These applications could be of help in the field since one of the major issues in studying HSCs metabolism is the limited sensitivity of the "standard" assays, which make them not suitable for HSC studies.

However, the observations do not fully support the claims. There are no direct evidence/experiments tackling cell cycle state and metabolism in HSCs. Often the observations for their claims are indirect, while key points on cell cycle state-metabolism, OCR analysis should be addressed directly.

Specifically, there are several major points that rise concerns about the claims:

1. The gating strategy to select HSCs with enlarged Sca1 gating is not convincing. I understand the rationale to have a sufficient number of cells to analyze, however this gating strategy should be applied also in the control group. From the FACS plot seems that there are more HSCs upon 5FU treatment (Figure S1b). How that is possible? Is it because of the 20% more of cycling cells at day 6? To prove that this gating strategy still represents a pure HSC population, authors should compare the blood reconstitution capability of this population with a "standard" gated population. If the starting population is highly heterogeneous then the metabolic readout could simply reflect cell heterogeneity.
2. S2 does not show major differences before and after sorting. However, a key metabolite like Lactate is decreased, which is also one of the most present. Wouldn't that mean that HSCs once they move out from the hypoxic niche, they decrease lactate production? Do they decrease anaerobic glycolysis? How can quiescent HSC mostly rely on OXPHOS being located in hypoxic niche?
3. The authors performed challenging experiments to track radiolabeled glucose, which are quite remarkable. However, the data do not fully support the conclusions. Mitochondrial metabolism in HSCs can be supported by fatty acid and glutamate, thus authors should track the fate of other energy sources to fully discriminate the glycolysis vs mito-metabolism dependency. From the data on S2 and Fig1 1C-F, the authors can conclude that upon 5FU treatment HSCs increase glycolytic rate.
4. In Figure S1, 5-FU leads to the induction of cycling HSCs and in figure 1, 5-FU results in higher activation of glycolysis. Would it be possible to correlate these two phenotypes together? For example, by sorting NBDG+ cells and checking the cell cycle status of these cells?
5. FIG.2B-C: Increase of Glycolysis upon oligomycin treatment is common in many different cell types. As explained before, other radiolabeled substrates should be used to understand the real effect on mitochondria metabolism.
6. Why are only ECAR measurements (and not OCR measurements) shown? In Fig.2G, why are HSCs compared with cKit+ myeloid progenitors, and not with MPP1? The ECAR increased

observed in HSC upon oligomycin treatment is shared with many other types of cells. However, cKit⁺ cells have a weird behavior. Upon oligo treatment cKit⁺ cells decrease ECAR, which is quite unusual. The data of both HSCs and cKit⁺ cells could be clarified by adding OCR curves. Moreover, it is recommended to run glycolysis stress test profile to assess the dependency to glycolysis (Glucose, Oligomycin, 2DG).

7. Since HSCs in the niche are located in hypoxic regions of the bone marrow, would that not mimic OxPhos inhibition (oligomycin)? Would that not mean that HSCs in the niche are more glycolytic (anaerobic glycolysis)?

FIG.3 A-C. As mentioned previously, the flux analyses should be integrated with data using other energy sources. If cycling HSCs are less dependent to OXPHOS, what happen if you inhibit OXPHOS in 5-FU condition? Since the authors are linking OXPHOS inhibition and upregulation of Glycolysis to increase proliferation, do HSCs proliferate more when treated with oligomycin?

8. FIG.4 shows that in vivo administration of radiolabeled glucose especially marks metabolites of TCA cycle and Glycolysis. The authors interpret enhanced anaerobic glycolysis, but I am not sure this is correct; if more glycolysis products go in the TCA cycle, it might mean that HSC start engaging mitochondrial metabolism. What do the authors think about that?

9. FIG.4: the experimental design is not clear. Are BMNNCs stained and then put in culture? Is it 6-day culture or BMNNCs are purified at day 6 post 5FU? FIG-4B-C The difference between PBS vs 5FU conditions are the most significant; however, the effect of oligomycin in both conditions is the most dramatic one. From this readout, it seems that HSCs are more dependent on mitochondria for energy production both upon 5FU treatment and in PBS conditions.

10. In Figure 5B, the orange line (Glucose+OXPHOS inhibition) remains stable, which means HSCs prefer to use glycolysis when OXPHOS is inhibited. Which metabolic pathway would HSCs use under hypoxic conditions? As HSCs resides in hypoxic niche, does it mean that these steady-state HSCs prefer to use glycolysis for ATP production? As mentioned before, mitochondrial inhibition can be comparable at the in vivo condition of the niche, where low pO₂ will "inhibit" mitochondria metabolism.

11. FIG.6H should be extended with cell cycle analyses. There are no differences between 5FU and ctrl groups. If 5FU induces HSCs cycling and increases glycolysis I would expect higher 2-NBDG uptake in the 5FU group. How do the authors explain this?

12. In S7 the experimental design is not clear. What are quiescent vs proliferative conditions? What does it mean "cell number of HSC-derived colony"? Is it a CFU assay? Then you should show colony numbers. When HSCs proliferate, they need more energy thus inhibition of metabolism will impact proliferation. What happens if you inhibit mitochondrial metabolism with oligomycin?

13. In FIG 7 since homing of HSCs is influenced by the cell cycle state, should be important to show if in the genetic model for PFKFB3 in HSCs there's a difference in homing efficiency.